

Sports game intervention aids executive function enhancement in children with autism - An fNIRS study

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ARTICLE INFO

Keywords:

Autism
Sports game
Executive function
Brain activation
Functional connection

ABSTRACT

Executive dysfunction is a prevalent issue in children diagnosed with autism spectrum disorder (ASD). While the efficacy of physical exercise in enhancing cognitive abilities in these children is well-documented, research exploring the relationship between physical exercise and brain function remains limited. This study aimed to investigate the impact of cognitively stimulating exercise on executive functions (EF) in children with ASD. The study enrolled thirty children with ASD who were randomly allocated into two groups: a sports game learning group ($n = 15$) and a control group ($n = 15$). Functional near-infrared spectroscopy was utilized to monitor cerebral function alterations pre- and post- an eight-week intervention program. The study focused on three core components of executive function: working memory, inhibitory control (IC), and cognitive flexibility (CF). Results revealed a significant improvement in the EF in the intervention group. After eight weeks of intervention, neural activity, along with improved EF performance, was enhanced significantly in the prefrontal cortex (PFC). During post-intervention, EF tasks were also significantly activated in the dorsolateral PFC, orbitofrontal cortex, and frontal pole area. Furthermore, an increase in short-distance functional connectivity within the PFC was observed during resting states. These results imply that engagement in sports game training can significantly improve EF information processing, augmenting task-related cortical activations and the efficiency of brain function networks in children with ASD.

1. Introduction

Autism spectrum disorder (ASD), a neurodevelopmental disorder, is primarily identified by deficits in social abilities, repetitive behaviors, and confined interests manifesting in early childhood [1]. Besides these core symptoms, executive dysfunction (EDF) emerges as a prominent neurodevelopmental characteristic of ASD across various developmental stages [2]. Notably, individuals with ASD exhibit significant cognitive impairments in tasks demanding an understanding of social-emotional contexts, thereby linking this deficiency to aspects of language development and social integration.

Executive function (EF) [3] plays a pivotal role in orchestrating final behaviors, which manifest as voluntarily planned actions. This is achieved through goal representation in working memory (WM) [4], the

suppression of automatic, task-inappropriate responses (inhibitory control, IC) [5], and the adaptive switching of task objectives (cognitive flexibility, CF) [6]. Numerous studies postulate that EDF in individuals with ASD stems from aberrations in the brain's system, impairing complex information processing [7,8]. In WM tasks, children with high-functioning ASD exhibit reduced functional connectivity in the left prefrontal-parietal circuit relative to typically developing children [9,10], and diminished connectivity between the prefrontal cortex (PFC) and posterior brain regions in IC [11] and CF task [12,13]. In other respect, research also reveals attenuated connectivity among visual, temporal, and parietal regions during psychological attribution tasks in ASD patients [14]. This culminates in a modular "cognitive-control network," wherein associated brain regions interact in a closely integrated manner, facilitating cognitive control within task contexts [15].

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<https://doi.org/10.1016/j.neulet.2024.137647>

Received 24 September 2023; Received in revised form 25 December 2023; Accepted 15 January 2024

Available online 17 January 2024

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In summary, cortical activation and functional connectivity between brain regions in children with ASD are responsive to top-down modifications in cognitive behavior.

Currently, integrated-treatment (computer-based cognitive, pharmacological, and strategy-based) is the main intervention strategy for improving EDF in children with ASD. While certain studies report enhancements in EF in ASD children following such interventions, their implementation is complex, necessitating specialized equipment and trained professionals [16,17]. These tools, despite their effectiveness in fostering EF skills development in a safe, interactive, and engaging manner for both children and adults, are prohibitively expensive. In contrast, sports interventions, recognized for their high effectiveness, low cost, absence of side effects, and ease of implementation, are increasingly being adopted for the rehabilitation of autistic children [18]. Numerous studies affirm the efficacy of physical activity in enhancing EF [19] and brain networks [20] in children with ASD. Transitioning from laboratory settings to more ecologically valid, naturalistic contexts, research emphasis has shifted to qualitative aspects of exercise interventions on executive functioning, such as the type of exercise. Here, motor skills, fundamental to physical activity, have garnered significant attention. The cognitive stimulation hypothesis, advocated by Best [21] and Pesce [22], posits that cognitively engaging exercises are more likely to yield cognitive improvements than those lacking cognitive stimulation. In animal studies, complex, randomized movements have been shown to more effectively induce neural neurogenesis in the cerebral cortex compared to simple, repetitive movements [23]. This stems from the notion that cognition emerges from diverse actions, including movement and interaction, with the body being central to cognitive activity. Cognitive experiences are often rooted in bodily experiences. The body and its motor system are integral to mental activities; cognition, thinking, emotion, and reasoning are all grounded in the body, whose interactions with the external environment shape the nature and content of mental activities [24]. Consequently, we have developed a sports game centered around cognitive exercise to intervene in the EF of children with ASD.

As research advances, it becomes increasingly imperative to investigate the underlying causes and mechanisms of this observed pattern. The current study aims to extend the understanding of the theoretical underpinnings of cognitive exercise and brain plasticity. Therefore, we hypothesize that engaging in sports game could facilitate changes in EF behaviors among these children by enhancing brain function. Furthermore, it seeks to offer novel insights and empirical support for the ongoing development and enhancement of intervention programs that leverage sport to augment executive functioning benefits in children with ASD.

2. Methodology

2.1. Participants

Employing the G*Power calculation, key parameters were established as follows: (1) a medium effect size (Cohen's $f = 0.25$), (2) a power of 0.85, (3) an alpha level of 0.05, and (4) a repeated measures analysis of variance encompassing two groups (experimental and control). This calculation indicated a requisite minimum sample size of 26 participants, with 13 individuals in each group, to achieve statistical significance. Participants were recruited from the Juxian Limin Hospital in Rizhao, China, comprising children diagnosed with ASD. The inclusion criteria were stringent: (1) Diagnosis of ASD in accordance with the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-V), validated through clinical assessment and the Autism Diagnostic Observation Schedule–2nd Edition by a qualified assessor; (2) Age range of 3 to 6 years and an IQ ≥ 70 ; (3) Absence of hearing and visual impairments; (4) No comorbid psychiatric diagnoses or neurological disorders, such as epilepsy; (5) Non-use of central nervous system drugs in the short term; (6) No history of cranial injury; (7) No head trauma,

physical disabilities, significant organ diseases, or substantial musculoskeletal injuries in the past two years; and (8) Approval from a medical supervisor for participation in the study.

2.2. Procedure and regulatory oversight

In this study, thirty children were randomly allocated to either the intervention or control group using a computer-generated randomization table (REDCap), designed by staff not directly involved in the study (refer to Fig. 1 for the flowchart). The participants were divided into two groups: (1) conventional therapy + learning sports games, and (2) conventional therapy. Pre-assessment and post-assessment were conducted for all participants. To mitigate participant fatigue, cognitive assessments were spread over two consecutive days, with each session not exceeding 30 min. Additionally, the sequence of cognitive assessments was standardized across the experiment to minimize potential order effects.

Cognitive Task: At the outset of the experiment, participants were seated in chairs identical to those used in their regular classroom settings. Following the calibration of fNIRS device, a 30-second quiet period was observed, succeeded by practice sessions for the tasks. Each task was rehearsed ten times prior to the commencement of the experiment. Upon completion of the practice phase, the researcher initiated by pressing the “Start” button. Participants began the task upon hearing the “start” prompt from the device and ceased upon the “Stop” command. The experiment was structured into eight blocks, each comprising a 30-second task duration interspersed with 10-second rest intervals.

Resting State: Throughout the experiment, subjects were seated comfortably in a tranquil, dimly-lit room. They were instructed to keep their eyes closed, remain alert, and minimize bodily movements as much as possible. Given that children with ASD may face challenges in maintaining head stillness, the duration for data acquisition was set at 2 min [25].

This study adhered to the principles outlined in the Declaration of Helsinki for all experiments and data analyses. Prior to commencing the experiment, guardians of the participating children and hospital physicians were thoroughly briefed about the basic procedures, precautions, and potential risks involved in the experiment. In addition, written informed consent was obtained from the foster guardians of the children, ensuring their understanding and agreement to the study's protocols.

2.3. Interventions

Experimental Group: This study meticulously selected sports games tailored for children with ASD, considering the appropriateness of program choices, implementation, and rewards. The chosen activities included athletics, basketball, and football games. Throughout these interventions, the activity intensity was maintained at a moderate level, with participants' mean heart rate controlled within the range of $(220 - \text{age}) \times (60\% - 69\%) = 128\text{--}148$ beats per minute. Heart rates were monitored using a Polar H10 monitor (Polar Electro, Inc., Finland). The intervention spanned eight weeks, with sessions conducted six times per week, each lasting 30 min [26,27]. Each session was divided into three phases: a 5-minute warm-up, a 20-minute game period, and a 5-minute relaxation phase. The selection of sports games (athletics, basketball, and football) was based on three primary aspects [28]: sporting characteristics, intensity, and sporting situations, all of which are instrumental in influencing EF through physical activity.

Control Group: The control group did not undergo sports games intervention. During the study period, the group members were asked to maintain their usual hospital intervention with no additional physical activity. At the end of the study, they were then given sports games intervention for eight weeks to recognize their contribution as a control group.

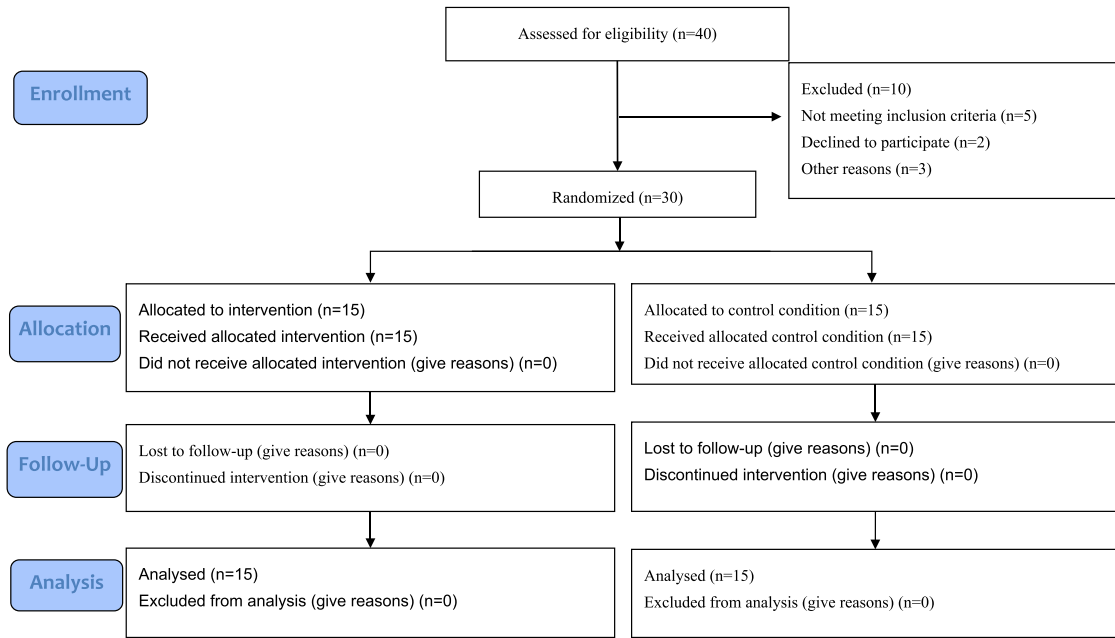


Fig. 1. CONSORT 2010 flow diagram.

2.4. Outcome measures

For cognitive assessment in this study, we employed validated experimental tasks. WM was measured using the Digital Span Task [29], IC was assessed with the Day/Night Task [30], and CF was gauged using the Dimensional Change Card Sorting [31]. Recognizing that participants might not initially comprehend the task requirements, it was necessary to clearly explain and practice the rules prior to the formal experiment until the subjects fully grasped the task's guidelines. To ensure consistency in cognitive engagement, ten pre-tests were conducted before initiating the official tasks.

2.5. fNIRS data acquisition

We utilized the latest generation of a portable fNIRS equipment (Nirxmart; Danyang Huichuang Medical Equipment Co., Ltd., Beijing, China) which consists of a hat, a Lenovo laptop, a near-infrared light source (emitting 730 and 850 nm), and a receiver detector. The sampling frequency was 10 Hz. The specific brain regions under investigation are depicted in Fig. 2, with the corresponding channels detailed in Table 1.

Table 1

Regions of interest and the channels used in the present study.

Region	Left brain channels	Right brain channels
BA	CH1, CH2, CH5, CH6,	CH19, CH20, CH21, CH22
DLPFC	CH3, CH7, CH8, CH10	CH14, CH15, CH17, CH18
OFC	CH4, CH11	CH12, CH16
FPA	CH9	CH13
M1	CH32, CH31	CH23, CH24
S1	CH33, CH34	CH25, CH26
SA	CH35, CH36	CH27, CH28
V3	CH37, CH38	CH29, CH30

2.6. Statistical analyses

The statistical analysis of the collected behavioral data was conducted using SPSS 25.0 software (SPSS Inc). To ascertain homogeneity between the experimental and control groups prior to the sports game intervention, the EFs scores and demographic variables of both sets of subjects were compared and analyzed using independent samples t-tests. Subsequently, the differences between the experimental and control groups were evaluated through repeated measures analysis.

The data collected through fNIRS were analyzed using NIRS-SPM, available at (https://www.nitrc.org/projects/nirs_spm/), within the

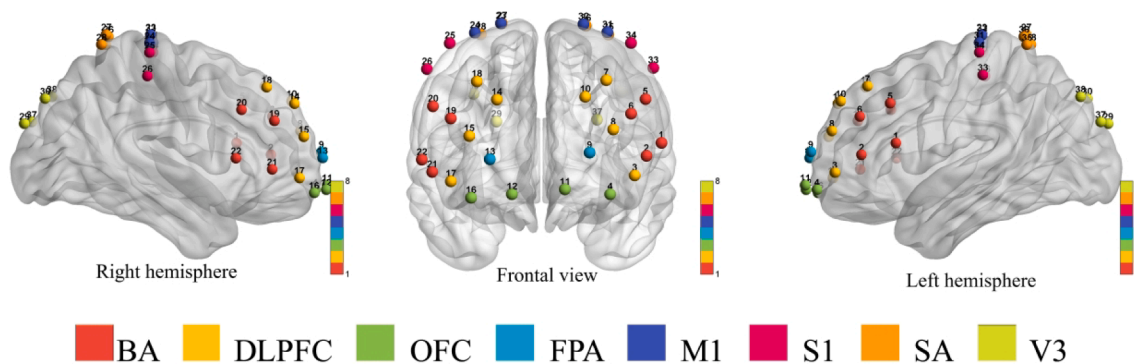


Fig. 2. fNIRS device channels correspond to the Broca's area (BA), the dorsolateral prefrontal cortex (DLPFC), the orbitofrontal area (OFC), the frontopolar area (FPA), the primary motor cortex (M1), the somatosensory cortex (S1), somatosensory association-cortex (SA), and the visual association cortex (V3).

Matlab (2014a) operating environment. Initially, the recorded light intensity signal was converted into a light density signal. Subsequently, Principal Components Analysis (PCA) was employed to eliminate physiological noise and motion artifacts, setting the PCA threshold percentage to 97 %. Further, to remove additional noise sources such as heartbeat and respiration, and to de-drift the data, we utilized the hemodynamic response functions (HRF) and the wavelet-minimum description length (Wavelet-MDL) analysis methods. The general linear model (GLM) was then applied to integrate task effects and evaluate the activation in relevant brain regions. Within the GLM framework, beta values were employed as comparative indicators for independent variables. These beta values provide insights into the direction (positive or negative) and magnitude of changes in hemoglobin oxygenation (indicative of cortical activity) for each condition. Finally, the beta values for each channel in both the experimental and control groups were analyzed using SPSS 25.0 for repeated measurements. The results were adjusted for the false discovery rate (FDR) to ensure statistical accuracy.

For the analysis of resting-state functional brain connectivity in children with ASD before and after the sports game intervention, the FC_NIRS toolbox, accessible at (<https://www.nitrc.org/project/s/fcnirs/>), was utilized. This process involved tracking the time series variations in the concentration of oxyhemoglobin for each channel, which reflected the functional characteristics of the corresponding brain region. Thresholding was conducted based on a sparsity (S) level of 20 %, resulting in a 38*38 correlation coefficient matrix derived through Pearson correlation calculations. The connectivity data across the 38 channels were subsequently visualized using the BrainNet Viewer, available at (<https://www.nitrc.org/projects/bnv/>). This visualization facilitated a more comprehensive understanding of the functional connectivity patterns in the brain of children with ASD, both pre- and post-intervention.

3. Results

3.1. Preliminary analyses

Independent variables such as age and BMI of the participants in both groups were examined using independent samples t-tests. The results of these tests confirmed that the two groups were homogeneous, as detailed in Table 2.

3.2. Changes in the performance of cognitive tasks pre- and post-intervention

The behavioral aspects of EF in children with ASD were analyzed using repeated measures analysis, both pre- and post-intervention. The analysis revealed a highly significant difference in the interaction effect for IC scores over time and between groups, with $F(1, 28) = 19.164$, $p =$

0.000, and $\eta^2 = 0.406$. Similarly, for CF scores, the interaction effect of time and group was statistically significant, as evidenced by $F(1, 28) = 11.024$, $p = 0.003$, and $\eta^2 = 0.300$. However, WM scores did not exhibit a significant interaction effect, with $F(1, 28) = 2.206$, $p = 0.149$, and $\eta^2 = 0.073$. In summary, the eight-week sports game intervention markedly improved the behavioral aspects of IC and CF, but not WM, in the EF of children with ASD, as detailed in Table 3.

3.3. Cortical activation pre- and post-intervention according to fNIRS results

A repeated measures ANOVA was employed to analyze the beta values from the fNIRS data, focusing on the interaction between the group (experimental vs. control) and time (pre- and post-intervention). The beta values served as the dependent variable, with the presence or absence of the sports game intervention as the between-group factor and the timing of the intervention as the within-group factor. Cluster analysis ($p < 0.05$) of these beta values also resulted in a cluster activation plot, showcasing the changes in both groups before and after the sports game intervention, as illustrated in Fig. 3.

For WM task, a significant group*time interaction was observed in specific channels. These channels corresponded to brain areas including L-BA ($p = 0.010$), L-DLPFC ($p = 0.036$), and R-DLPFC ($p = 0.022$), as shown in Fig. 3(C).

During IC task, significant interactions were found in channels corresponding to L-DLPFC ($p = 0.002$), L-FPA ($p = 0.004$), R-FPA ($p = 0.031$), right dorsolateral prefrontal cortex R-DLPFC ($p = 0.019$), R-OFC ($p = 0.028$), and R-V3, ($p = 0.018$), as depicted in Fig. 3(A).

For CF task, channels showing significant group*time interactions included those linked to R-BA ($p = 0.049$), L-FPA ($p = 0.028$), L-DLPFC ($p = 0.006$), R-DLPFC ($p = 0.009$), and L-V3 ($p = 0.040$), as illustrated in Fig. 3(B).

It is noteworthy that the area of activation in cognition-related brain regions increased significantly in the experimental group following the sports game intervention.

3.4. Functional connectivity pre- and post-intervention during the resting state

Fig. 4 illustrates the changes in functional connectivity among various brain regions before and after the intervention. Notably, there was a highly significant enhancement in the connectivity between L-DLPFC and R-DLPFC. Additionally, significant enhancements were observed in the connectivity between the L-DLPFC and L-FPA, between L-OFC and L-PFA, and between L-M1 and R-M1. These findings indicate notable improvements in neural communication across these regions following the intervention.

3.5. Correlation of behavioral and brain activation changes

In the experimental group, the correlation between brain region activation and behavioral outcomes in children with ASD was examined using changes in T values derived from SPM8 first-order analyses, alongside alterations in sub-functional scores of EF, as depicted in Fig. 5. A significant positive correlation was observed between the activation in L-DLPFC and the behavioral outcomes in IC task ($r = 0.600$, $p = 0.018$). Additionally, there was a significant positive correlation between the activation in L-FPA and the behavioral outcomes in CF task ($r = 0.536$, $p = 0.039$). These findings suggest a strong relationship between the activation in specific brain regions and improvements in corresponding behavioral tasks among children with ASD.

4. Discussion

The primary objective of our study was to ascertain the impact of sports game on cognitive functioning in children with ASD. We posited

Table 2
Demographic of participants (M $\pm$ SD).

	Sport games group(n = 15)	Control group(n = 15)	P
Males	12	13	
Females	3	2	
Age (yr)	5.27 $\pm$ 0.70	5.07 $\pm$ 1.10	0.558
BMI (kg/m ²)	17.48 $\pm$ 1.65	16.86 $\pm$ 2.52	0.433
RBS-R	29.47 $\pm$ 6.22	29.20 $\pm$ 7.18	0.914
Narrow-minded behavior	3.27 $\pm$ 1.71	2.47 $\pm$ 1.68	0.207
Aggressive behavior	8.60 $\pm$ 1.96	9.00 $\pm$ 2.27	0.609
Self-injurious behaviors	5.20 $\pm$ 1.61	4.93 $\pm$ 1.67	0.565
Compulsive behavior	4.33 $\pm$ 1.59	3.20 $\pm$ 1.42	0.051
Ritual behavior	3.07 $\pm$ 1.49	4.20 $\pm$ 1.74	0.065

Abbreviations: M = mean; P = P-value; SD = standard deviation.

Table 3
Comparison of behavioral performance on cognitive tasks in the sports game and the control groups (score) (M ± SD).

	Sports game(n = 15)		Control (n = 15)		Time <i>p</i> -value (<i>F</i> -value)	Group <i>p</i> -value (<i>F</i> -value)	Time*group interaction <i>p</i> -value <i>F</i> -value, η^2
	Pre-	Post-	Pre-	Post-			
WM	6.7 ± 4.8	7.3 ± 4.7	6.8 ± 4.5	6.9 ± 4.1	0.080, <i>F</i> (1,28) = 3.296	0.920 <i>F</i> (1,28) = 0.010	0.149 <i>F</i> (1,28) = 2.206 η^2 = 0.073
IC	8.4 ± 5.0	10.7 ± 4.1	9.3 ± 4.7	8.3 ± 4.1	0.101, <i>F</i> (1,28) = 2.881	0.657, <i>F</i> (1,28) = 0.201	0.000 <i>F</i> (1,28)=19.164 η^2 = 0.406
CF	6.0 ± 6.4	7.1 ± 4.7	9.1 ± 3.7	8.9 ± 3.7	0.056, <i>F</i> (1,28) = 3.969	0.186 <i>F</i> (1,28) = 1.842	0.003 <i>F</i> (1,28) = 11.024 η^2 = 0.300

Abbreviations: M = mean; SD = standard deviation; Pre- = Pre-intervention; Post- = Post-intervention.

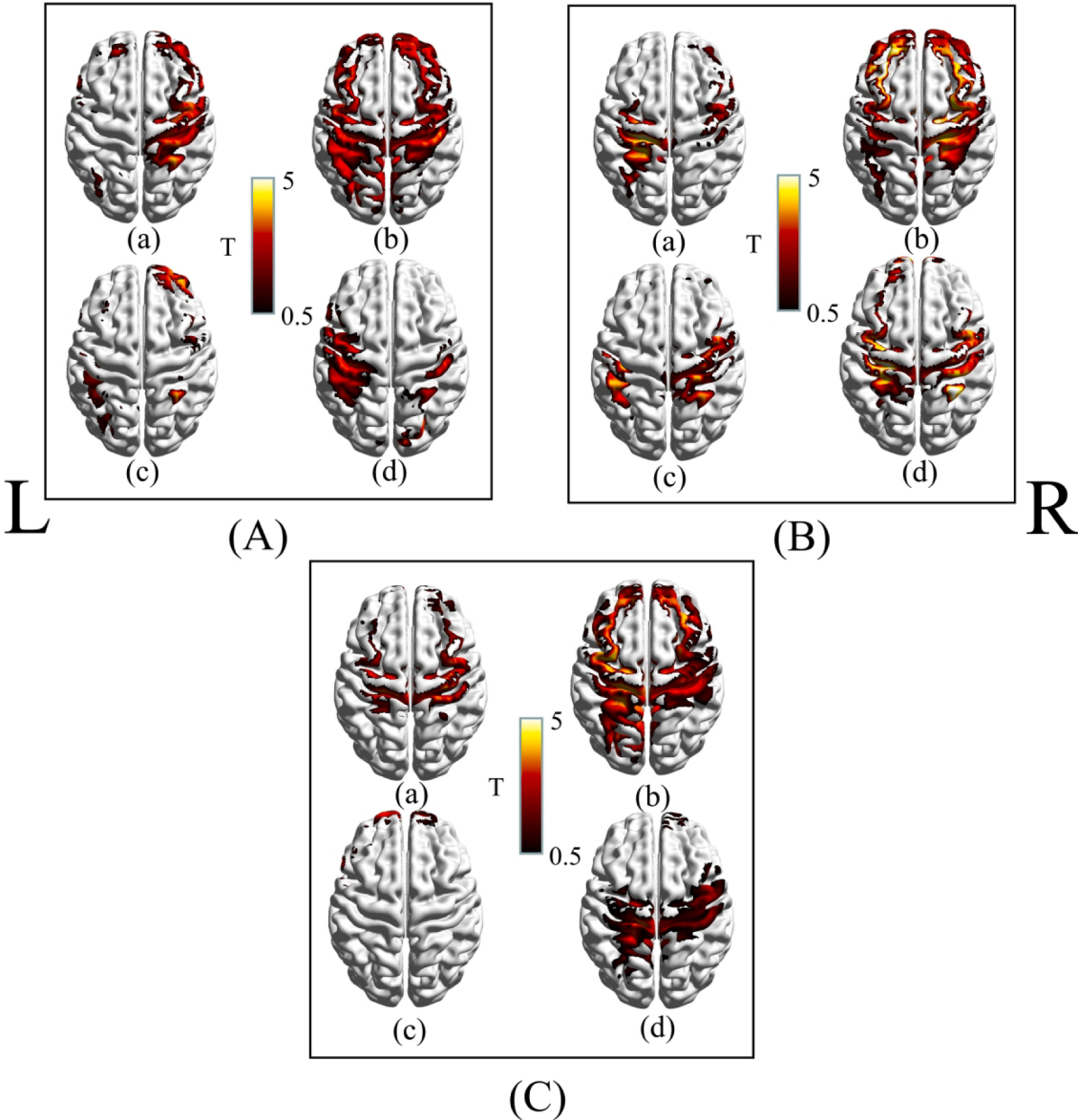


Fig. 3. Cohort activation plots of IC in the experimental and control groups before and after the sports game intervention: (A) IC test; (B) CF test; and (C) WM test. Note: (a) Pre test for the experimental group; (b) Post test for the experimental group; (c) Pre test for the control group; and (d) Post test for the control group, (*p*-value < 0.05).

that engaging in sports game could facilitate changes in EF behaviors among these children by enhancing brain function, thereby exerting a top-down influence. The findings of our study partially corroborated these hypotheses.

4.1. Effects of sports game on brain activation in children with autism

Post-intervention, significant activation of DLPFC was observed during WM tasks, as indicated by fNIRS data. This finding aligns with McNab et al. [32], who noted that DLPFC and parietal cortex activation correlates with the amount of information involved in memory tasks,

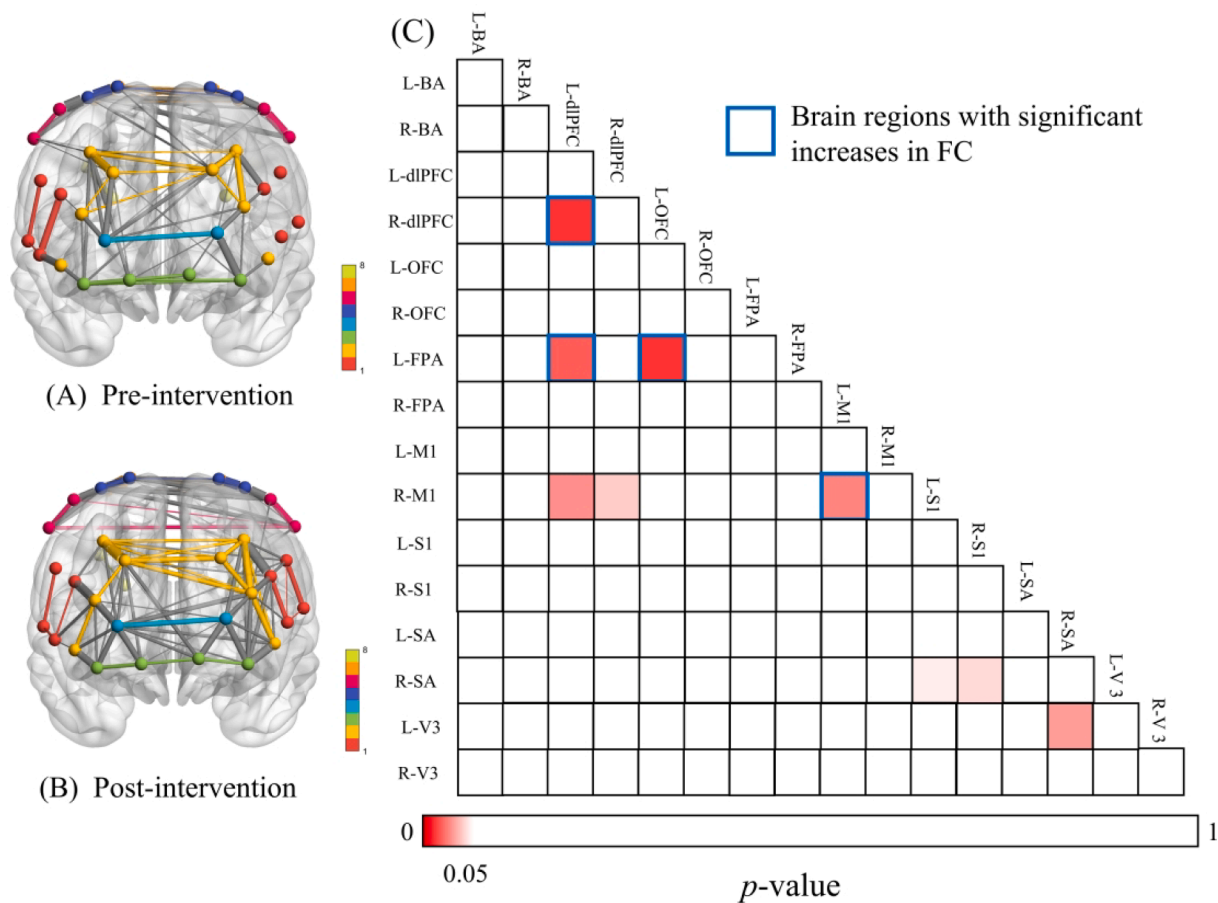


Fig. 4. Functional connectivity of brain areas before and after the eight-week sports game intervention: (A) Functional connectivity of individual brain areas before the sports game intervention; (B) Functional connectivity of individual brain areas after the sports game intervention; and (C) Individual brain areas with significant changes in functional connectivity before and after the sports game intervention.

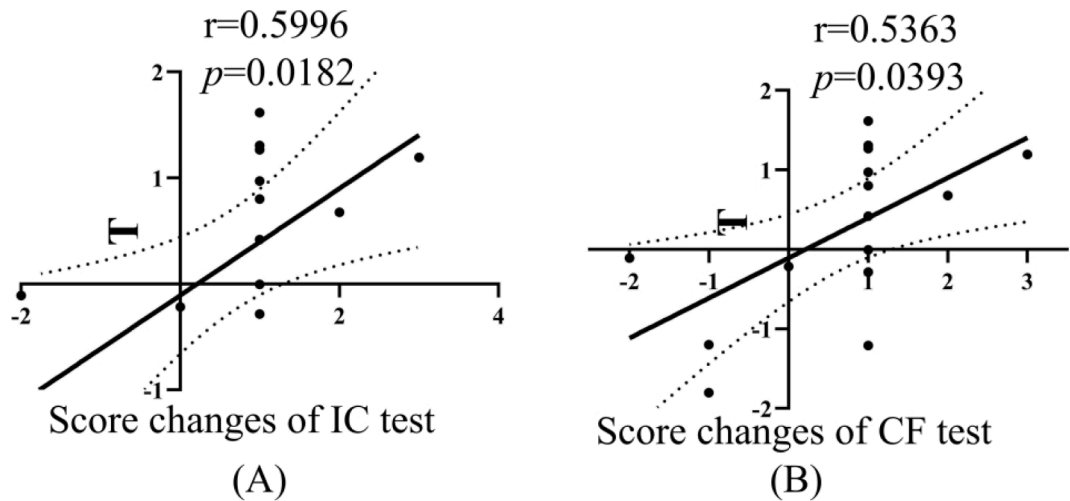


Fig. 5. Correlation between changes in brain activation and in executive function in children with ASD because of sports game intervention: (A) Relationship between the T-values of L-DLPFC and changes in IC scores and (B) Relationship between the T-values of L-FPA and changes in CF scores.

and differences in WM capacity are linked to activation in these regions. WM tasks prompt a reciprocal allocation of cognitive resources in the PFC areas of the cerebral cortex, leading to changes in oxyhemoglobin concentration and brain area activation. While both DLPFC and parietal cortex are responsive to WM tasks, their effects differ between regions. The DLPFC is specifically involved in encoding and manipulating

information in brain regions during WM tasks [33]. Therefore, the sports game in the Digital + Game genre, emphasizing multivariate movement imitation and transient digital memory, can enhance DLPFC activation. In the IC task, significant activation was noted in inhibitory-related brain regions, especially in the DLPFC, FPA, and R-OFC, with L-DLPFC and L-FPA showing pronounced activation. These changes are attributed

to the interplay between abundant inhibitory neurons in the DLPFC and FPA and the excitatory neurons in the anterior cingulate gyrus [34]. The 8-week intervention likely strengthened this interplay, enhancing the behavior of children with ASD from a top-down approach. Fig. 5(A) illustrates the correlation between L-DLPFC activation and improved behavioral scores. Additionally, while significant R-DLPFC activation is typically observed in IC tasks in normally developing children [35], our study found significant activation in multiple brain areas in children with ASD post-intervention, indicating a compensatory phenomenon [36]. This suggests that children with ASD have comparatively poorer IC functioning. A diverse and varied sports game, focusing on inhibiting dominant movements and adapting to new roles, can thus enhance the activation of the DLPFC and FPA, improving IC circuits in children with ASD.

During the CF task, significant activation was observed in the DLPFC, L-FPA, and L-V3, with DLPFC showing substantial significance. Prior studies [37,38] have linked the PFC, FPA, cingulate gyrus, and cerebellum to CF in children at rest, as observed through functional MRI. Fig. 5(B) further indicates that L-FPA activation positively correlates with behavioral changes in CF tasks, suggesting an enrichment of the cerebral cortex areas associated with CF. Hence, a diverse and varied sports game, designed around the concept of switching motor tasks and hand-eye coordination, can improve the activation of the DLPFC and consequently enhance CF in children with ASD.

4.2. Effects of sports game on resting-state functional connectivity of children with autism

The study revealed a marked increase in functional connectivity during the post-intervention resting state compared to the pre-intervention state. These findings are consistent with previous research [44] which indicates that functional connectivity in childhood is primarily characterized by enhanced local connectivity, evolving to include middle and long-distance regions post-adolescence. Consequently, we observed that cognitive-related regions (DLPFC, FPA, OFC) exhibited increased interconnectivity in the experimental group, suggesting that the intervention improved the cognitive region network in children with ASD. Considering that all participants were right-handed, the study also explored the possibility of a lateralized physiological response in relation to functional connectivity. The findings showed more significant improvement in the left regions as compared to the right, post-intervention. This may be attributed to the left hemisphere's dominance over the actions and organs on the right side of the body [45], although this hypothesis warrants further investigation. In summary, the sports game intervention effectively optimized the brain network in children with ASD, particularly by enhancing connectivity between various regions associated with cognition.

4.3. Effects of sports game on executive functioning in children with autism

This study, guided by the Multiple Pathways Theory of Physical Activity for the Development of Children's Brain Intelligence [46], implemented an 8-week sports game intervention program specifically designed for children with ASD. The program's structure required the children to learn new and complex motor skills within a dynamic group environment, necessitating engagement across various cognitive dimensions. This approach involved operational commands that were continually updated, prompting children with ASD to constantly refresh their working memory and integrate both old and new information, thereby fostering improvements in their executive functioning. In support of this approach, Moreau [47] discovered that the purposeful learning of complex, novel exercises had a more pronounced effect on cognitive enhancement compared to repetitive exercise. This finding underscores the crucial role of cognitive activity in augmenting cognitive performance through exercise. Furthermore, the concept of

embodied cognition [48] posits that cognition is a dynamic construct rooted in the body and time, inseparable from the body's dominance. Hence, physical activity acts as a conduit, linking cognition with the complex external environment and ensuring the effective development of cognitive abilities through physical practice.

5. Conclusions

The current research demonstrates that cognitive-based exercise exerts a positive impact on EF in children diagnosed with ASD. More precisely, enhancements in brain activation and functional connectivity foster improvements in cognitive behaviors from a holistic perspective. This study not only elucidates the neural mechanisms underlying the improvement of EF through sports games but also offers theoretical underpinning and guidance for the formulation of future rehabilitative exercise programs targeting EF in children with ASD.

CRedit authorship contribution statement

Hao Chen: Writing – original draft. **Qi Liang:** Formal analysis. **Baijie Wang:** Resources. **Hexiang Liu:** Data curation. **Guijun Dong:** . **Kefeng Li:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

This study was funded by the Humanities and Social Science Fund of Ministry of Education of China (Grant No. 23YJAZH026), the Project of Competitive Science and Technology Tackling Program of Quzhou (Grant No. 2023K239), the Project of Competitive Science and Technology Tackling Program of Quzhou (Grant No. 2023K217), the Shandong Judicial Administrative Drug Rehabilitation and Health Integration Pilot Project (Grant No. SDTWRH2021086), and the Integrated physical Health pilot Project for Autistic Children (Grant No. 2021031). The authors gratefully thank all the children who participated in this study, parents of children for their supports, and research assistants who helped with data collection and other contributions.

Ethics statement

The project was approved by the Medical Ethics Committee of the Quzhou College of Technology.

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